Digital Communications and Networks 12 (2026) 495–504

Contents lists available at ScienceDirect

Digital Communications and Networks

journal homepage: www.keaipublishing.com/dcan

RADAR: A rapid discovery algorithm for routes in SDN-based IoV ☆

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ARTICLE INFO

Keywords:
Route discovery
Software defined networking (SDN)
Link failure
Intra-zone communications
Inter-zone communications
*Theta**

ABSTRACT

Link failures are a critical issue that disrupts consistent communications in the fast-evolving Internet-of-Vehicles (*IoV*) environment. In response to the aforementioned limitation, we introduce the rapid discovery algorithm for routes (*RADAR*), a novel scheme designed for rapid route discovery in the event of abrupt link failures or topological changes. Built on a heuristic *Theta** algorithm, *RADAR* operates within a three-layered Software-Defined Networking (*SDN*) architecture to ensure seamless communications. The first layer collects vital vehicle information, including location, speed, direction, and vehicle *ID*. The second layer uses an *SDN* edge controller to carry out local and intra-zone path discovery with the *Theta** heuristic algorithm. The third layer introduces a global controller that oversees inter-zone route discovery, thereby expanding the network's reach and capabilities. To validate *RADAR*'s effectiveness, we compare its performance to those of *Dijkstra*-based and *A** path planning algorithms across key metrics—Route Discovery Messages (*RDM*), Path Length (*PL*), and Route Discovery Time (*RDT*). Our analysis shows significant improvements with *RADAR* over *Dijkstra*, achieving up to an 80% increase in *RDM*, 20% in *PL*, and 70% in *RDT*. It also surpasses *A** by up to 70% in *RDM* and 60% in *RDT*. These results indicate that *RADAR* enhances routing reliability in *IoV* systems, thereby improving overall network performance.

1. Introduction

The Internet of Vehicles (*IoV*) has emerged as a transformative paradigm for future transportation systems, enabling vehicles to interact intelligently with each other and the surrounding infrastructure. The highly dynamic nature of *IoV*, characterized by frequent topological changes and link failures, demands robust and efficient routing schemes to maintain uninterrupted communications [1]. To address these challenges, Software-Defined Networking (*SDN*) offers a compelling solution. By allowing programmable control and efficient management of network resources, *SDN* provides the flexibility and agility required to adapt to the volatile conditions inherent to the *IoV* environment. Integrating *SDN* with *IoV* promises to revolutionize the routing mechanisms, making them more reliable, rapid, and responsive to real-world conditions [2].

1.1. Background work

Numerous routing schemes have been proposed in the literature to address the complexities of *IoV*'s constantly evolving topology [3–5]. To maintain consistent and uninterrupted communications, *IoV* requires a routing strategy that either preserves connectivity over extended periods or rapidly re-establishes connections by identifying an alternative route in the event of a sudden failure. Previous work has highlighted the frequent communication link breakages in Vehicular Ad-hoc Networks (*VANETs*) due to high-speed vehicles.

In the context of ongoing advancements in *VANETs*, the integration of *SDN* is a notable development [6]. The ability of *SDN* to reconfigure dynamically and manage network resources globally has made it a potent solution to the inherent challenges of *VANETs*, such as frequent disconnections and the management of node mobility. The centralization inherent in *SDN* provides framework for enhancing scalability and improving routing decisions beyond the capabilities of traditional network models. Empirical studies suggest that adopting an *SDN* approach can significantly mitigate issues related to network congestion and the absence of fixed infrastructures, leading to a more robust and efficient networking environment [7,8]. Moreover, the predictive capabilities of

☆ Peer review under the responsibility of the Chongqing University of Posts and Telecommunications.

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https://doi.org/10.1016/j.dcan.2024.08.001
Received 26 December 2023; Received in revised form 22 July 2024; Accepted 2 August 2024

Available online 8 August 2024

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SDN, rooted in its global view of network topology, facilitate accurate forecasting of vehicle locations and trajectories, thereby minimizing the routing disruptions caused by vehicular movements [9]. This predictive feature also contributes to a decrease in the overall network overheads, streamlining the networking process within *VANETs*.

1.2. Motivations and limitations

Researchers have developed a novel Cluster-based *VANET*-oriented Evolving Graph (*CVoEG*) model [10], which extends the existing *VoEG* framework [11] to enhance vehicular communication reliability. The *CVoEG* model employs link reliability as a selection criterion for Cluster Members (*CMs*) and Cluster Heads (*CHs*). The eigengap heuristic divides *VANET* nodes, i.e., vehicles, into an Optimal Number of Clusters (*ONC*). Building upon the *CVoEG* model, a robust routing scheme named *CEG-RAODV* was introduced to identify the Most Reliable Journey (*MRJ*) from the source to the destination. However, the *CVoEG* model has its limitations. It is confined to a specific region covered by a single Road-Side Unit (*RSU*) and employs *CEG-Dijkstra* for route discovery, generating many beacon messages. These constraints geographically limit the model to a specific zone and increase its computational complexity due to the high volume of messages. Our proposed RApid Discovery Algorithm for Routes (*RADAR*) builds upon the foundation established by *CVoEG*, utilizing *VoEG* as the underlying structure for both algorithms. The key distinction lies in the integration of *SDN*, which *RADAR* to enhance communication both within (intra-zone) and across (inter-zone) *VANETs* (discussed in the contribution part). Furthermore, while *CVoEG* relies on the *CEG-Dijkstra* algorithm for pathfinding, *RADAR* advances the methodology by employing the $Theta^*$ algorithm, which brings a heuristic approach to route discovery in grid-based road environments.

Various cluster-based models like connectivity-based clustering prioritize link connectivity for group formation and *CH* selection [12]. Vehicular to Everything (*V2X*)-based two-level clustering schemes [13] utilize fuzzy logic algorithms for Level-1 *CH* (*L1CHs*) selection, factoring in variables like relative velocity and link reliability. Multi-hop Moving Zone (*MMZ*) clustering [14] fuses *IEEE 802.11p* with *5G* technology to minimize cellular hand-off costs, and Cellular-*V2X* (*C-V2X*) [15] employs multi-factorial criteria for zonal head selection. Most of these schemes rely on non-heuristic algorithms like *Dijkstra* for route discovery, which often results in high computational complexity and inefficiency due to excessive message exchanges [10,11,13]. Such limitations make these algorithms less scalable and not ideally suited for the dynamic environments typical in *IoV*. These drawbacks motivate our *RADAR* scheme, which aims to provide a more scalable, efficient, and rapid routing solution through heuristic approaches.

Several existing *SDN*-based routing schemes in vehicular networks exhibit certain limitations that our *RADAR* approach aims to address. For instance, the SDN enabled approach for vehicle path-planning (*SEARCH*) focuses on a complex 3 tier architecture integrating *UAVs*, *VANETs*, and *5G* cellular systems to enhance situational awareness and road conditions [16]. While innovative, this approach is highly dependent on multiple technologies and is not specifically designed for rapid route rediscovery in dynamic topologies. Similarly, *ROAMER*, and its upgraded version *SURFER*, leverage *RSUs* and combine geographical and carry-and-forward routing methods [17]. Although these schemes incorporate a distributed *SDN* architecture and even a Blockchain system for security, they do not employ heuristic algorithms for efficient and quick route discovery. Their reliance on non-heuristic methods like *Dijkstra's* algorithm adds computational complexity and latency, which can be detrimental in highly dynamic environments like *IoV*-based *VANETs*.

These existing approaches illustrate the need for a more focused, less complex, and rapid route discovery mechanism, specifically in dynamic vehicular networks. *RADAR* addresses these gaps by using a heuristic-based $Theta^*$ approach and operates on both intra-zone and inter-zone discoveries for more efficient route planning.

1.3. Contributions

In response to the limitations of existing work, we introduce *RADAR*, a novel scheme designed for rapid route discovery in the event of abrupt link failures or topological changes. Our architecture operates within an *SDN*-based *IoV* environment and consists of 3 layers. The first layer collects vehicle information, such as location, speed, direction, and vehicle *ID*. The second layer incorporates an *SDN* Edge controller that employs the $Theta^*$ heuristic algorithm [18] for local and intra-zone path discovery. $Theta^*$ is indeed an advancement over the traditional A^* algorithm. It extends the capabilities of A^* by incorporating Line-Of-Sight (*LOS*) checks in the pathfinding process, allowing for a more direct path planning that considers the real-world possibility of diagonal shortcuts, which are not typically in the grid-based approach of A^*. This results in $Theta^*$ often finding shorter and more realistic paths than A^*. The third layer expands the network's reach by introducing inter-zone discovery via a global controller, which maintains comprehensive routing information across all connected *SDN* Edge controllers.

The key contributions of *RADAR* include:

- Proposing an innovative architecture integrating *SDN* and *IoV* to accelerate vehicular communication processing speed, ensuring uninterrupted data flows.
- Utilizing the efficient $Theta^*$ heuristic for quick and reliable path discovery, measured against link reliability as a key metric for route selection.
- Introducing a versatile route discovery mechanism that operates in both intra-zone and inter-zone environments. The use of a global controller for inter-zone discovery significantly broadens the network coverage, setting *RADAR* apart from existing schemes.
- Conducting a comprehensive evaluation and comparison of *RADAR* with state-of-the-art routing schemes, specifically *Dijkstra* and A^*, chosen for their similarities to our $Theta^*$-based approach [18]. This allows for a fair assessment of *RADAR's* advantages and efficiencies.

The remainder of this paper is structured as follows: Section 2 delves into the integration of *SDN* and *IoV*, as well as system modeling. Section 3 provides an overview of *RADAR* and outlines its methodology. Section 4 presents and discusses the results, while Section 5 offers concluding remarks.

2. *SDN-IoV* integration: system modeling

2.1. Components

The proposed network architecture comprises various essential components, as illustrated in Fig. 1. These include vehicles equipped with On-Board Units (*OBUs*) and *GPS* modules for communications and location tracking, *5G*-enabled *RSUs*, zonal *SDN* edge controllers, a centralized global controller, and physical obstacles that impact vehicular communications [19].

Physical Obstacles: Physical obstacles such as buildings, trees, billboards, and other vehicles can significantly impact signal strength and data transmission rates. Understanding the characteristics and positions of these obstacles is crucial for optimizing the vehicular communication path and overall network reliability.

***SDN* Infrastructure:** The cornerstone of our proposed architecture is the integration of *SDN* technology, a paradigm that provides unparalleled adaptability and scalability. In the context of our architecture, the zonal *SDN* edge controllers serve as a crucial component that orchestrates the vehicular network's localizing operations [20]. The zonal *SDN* edge controllers have multifaceted roles, including but not limited to, data collection and real-time route discovery. They receive vehicle activity data through beacon messages sent via *5G*-enabled *RSUs*. These

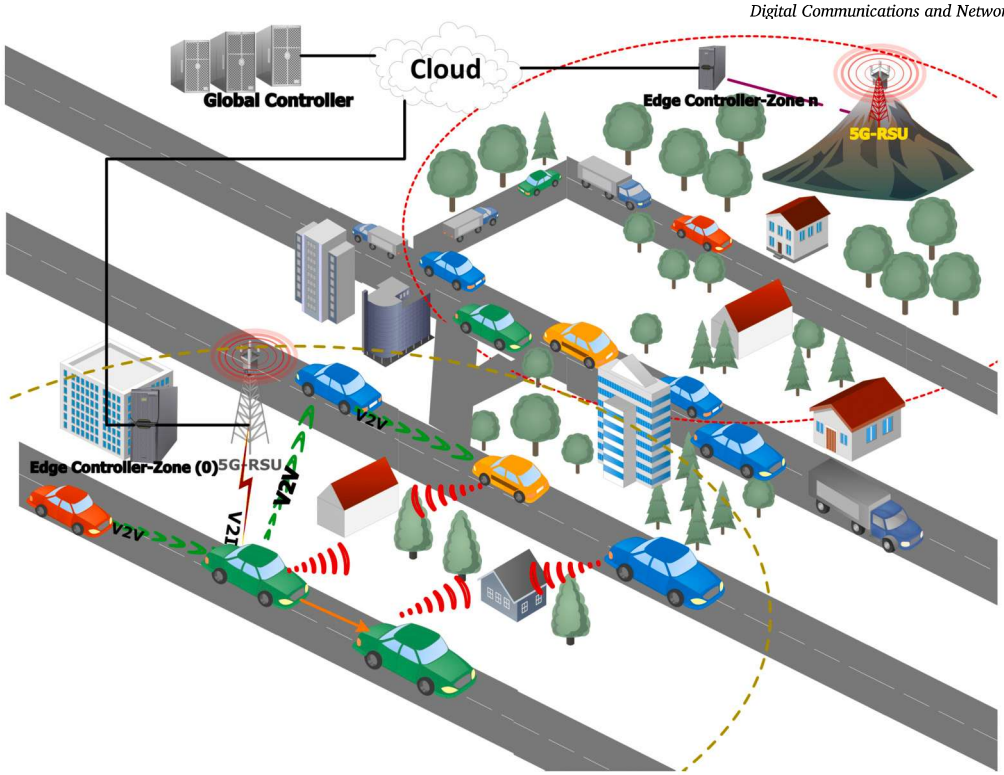


Fig. 1. A three-tier architecture of *RADAR* in *SDN*-based *IoV*.

beacon messages contain valuable information, such as the vehicle's current location, speed, and direction.

Zone-based Network Architecture: The road network is segmented into distinct zones, each defined by the transmission range of a specific *RSU*. Vehicles within these zones share data with their respective *RSUs*. Then, the data is relayed to the zonal edge controller for localized link and route discovery. When route discovery needs to extend beyond a given zone, the local edge controller coordinates with the centralized global controller to acquire a broader network perspective [21].

2.2. Link reliability model

In vehicular networks, the reliability of communication links between vehicles is pivotal for maintaining ongoing connectivity. A multitude of factors influence the availability and longevity of these links. These factors commonly include but are not limited to the signal strength, interference, vehicle density, speed of vehicles, distance between them, and even presence of physical obstacles such as buildings or trees [22]. Various researchers have previously addressed this challenge by introducing diverse parameters aimed at extending the operational lifetime of vehicle-to-vehicle links. For example, Malek et al. (2022) [23] focused on signal-to-noise ratios, while [24] explored the impact of vehicular density on link availability. In our proposed model, the link reliability concept has been used as a critical parameter for extending the effective lifetime of vehicle connectivity.

Definition 1 *(Link reliability)*. In the context of *VANETs*, the term *Link Reliability* is defined as the statistical probability that a direct communication link between two distinct vehicles, denoted as V_i and V_j, will remain continuously available over a specified time interval t.

Link reliability serves as a critical metric for evaluating the effectiveness and robustness of data exchange mechanisms within the network [25]. By carefully considering link reliability, our framework aims to achieve robust and uninterrupted communications among vehicles, thereby fulfilling the critical need for reliable data exchanges in *IoV*.

Let l be the communication link between two neighboring vehicles, and let R_T represent the combined transmission range of both vehicles. The duration D_l for which this link exists can then be defined as [10].

$$D_l = \frac{R_T}{\Delta V} \tag{1}$$

Here, ΔV stands for the differential velocity between the 2 vehicles and is calculated as $\Delta V = |v_i - v_j|$. The combined transmission range R_T for vehicles V_i and V_j moving in the same direction can be expressed as:

$$R_T = \begin{cases} r + d & \text{if } v_j < v_i \\ r - d & \text{if } v_i < v_j \end{cases} \tag{2}$$

In Equation (2), r denotes the transmission distance, and d is the Euclidean distance between the 2 vehicles. If the vehicles are moving in opposite directions, R_T is:

$$R_T = \begin{cases} r - d & \text{if } V_i \text{ and } V_j \text{ are moving apart} \\ r + d & \text{if } V_i \text{ and } V_j \text{ are moving toward each other} \end{cases} \tag{3}$$

Assuming vehicular speeds are normally distributed, duration T of communications between neighboring vehicles is also normally distributed [26]. The probability density function (pdf) of T is:

$$g(T) = \frac{R_T}{\sigma_{\Delta V}\sqrt{2\pi}T^2} e^{-\frac{(R_T/T - \mu_{\Delta V})^2}{2\sigma_{\Delta V}^2}} \tag{4}$$

Here, $\sigma_{\Delta V}$ and $\mu_{\Delta V}$ denote the standard deviation and the mean of the relative velocities, respectively. To evaluate $R_l(t)$, the reliability of link l at time t, we integrate pdf $g(T)$ over duration D_l as:

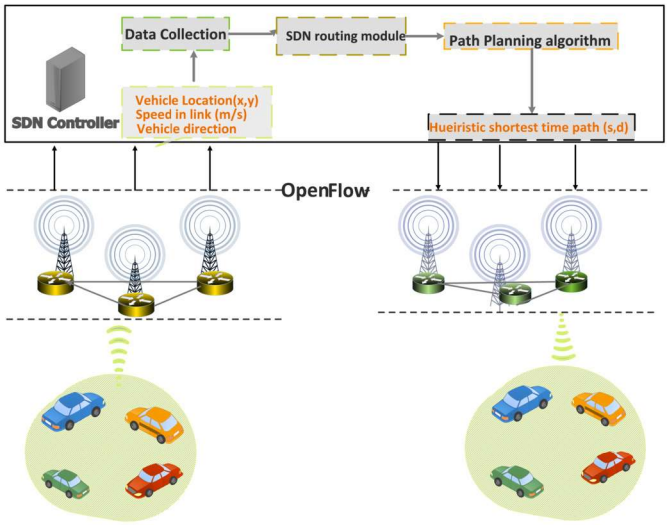


Fig. 2. Route discovery mechanism in *SDN*-based IoV controller layer architecture.

$$R_l(t) = \int_{t}^{t+D_l} g(T)\,dt \quad \text{if } D_l > 0 \tag{5}$$

Equation(5) quantifies reliability $R_l(t)$ of an individual link. A heuristic algorithm, i.e., $Theta^*$ is used by the edge controller to identify the reliable route for data packet transmission between the source and the destination [18].

2.3. Route discovery model

This section provides a detailed examination of the route discovery process within our proposed system. It outlines a structured approach across 3 distinct layers: the vehicle layer, the edge controller, and the global controller, each playing a pivotal role in efficient route identification.

2.3.1. Layer 1: vehicle layer

The vehicle layer serves as the cornerstone for initiating network communications, as shown in Fig. 2. Vehicles in this layer are responsible for consistently broadcasting beacon messages that carry crucial information, e.g., location, speed, and direction to *RSUs* within their transmission range. This layer sets the stage for the vehicular network, providing fundamental building blocks upon which higher-level functionalities are realized [27]. A distinct zone-based partitioning strategy is employed, wherein the operational area of each *SDN* edge Controller is segmented into specific zones for effective localized decision-making.

2.3.2. Layer 2: SDN edge controller

The *SDN* edge controller represents the second tier in our architecture and is pivotal for real-time route optimization (see Fig. 2) [28]. Continuously updated by *RSUs*, it employs our proposed *RADAR* to dynamically discover or reconfigure optimal routes, particularly in situations like new vehicle entries or link failures. Physical obstacles are also taken into account when making routing decisions. Each edge controller oversees a designated area that we refer to as a zone, which typically contains 2-3 *RSUs*. The controller maintains a comprehensive routing table for each vehicle within its jurisdiction and tracks neighboring vehicles. Given the volatile nature of *VANETs*, the speed of route discovery is crucial. The edge controller is optimized to manage this volatility efficiently, ensuring network stability and efficiency.

2.3.3. Layer 3: global SDN controller

Distinct from conventional *IoV* schemes limited to fixed geographic footprints, our approach significantly broadens network coverage. While edge controllers are designated for intra-zone communications, the global *SDN* server in the third tier of our architecture coordinates broader network activities. It specializes in inter-zone communications, collating route information from all Edge Controllers to build a cohesive and expansive routing strategy. This global perspective enables the network to adapt to various scenarios, including vehicles moving across multiple zones [29].

3. *RADAR*: RApid discovery algorithm for routes

3.1. RADAR overview

This section introduces *RADAR*, a heuristic algorithm engineered to rapidly and efficiently discover links and routes in *SDN*-based *IoV*. Utilizing a grid-based representation of road networks, *RADAR* can account for physical obstructions such as buildings and trees that may impede vehicular communications. The algorithm extends beyond traditional grid-based movements, and consisting of up, down, left, and right, to include diagonal directions [30]. This extended functionality is illustrated in Fig. 3. Additionally, *RADAR* incorporates a rapid recovery mechanism that activates upon link failure to identify alternative routes instantaneously.

A cornerstone of *RADAR* is its integration of the $Theta^*$ heuristic algorithm [18] into the *SDN*-based *IoV* environment. $Theta^*$ improves on conventional pathfinding algorithms by allowing for more natural and optimized routes that are not confined to grid-based movements. It also includes LOS checks between nodes, enhancing routing efficiency. This feature is vital in dynamically changing vehicular networks [31]. The application of $Theta^*$ within *RADAR* serves two primary functions:

- **Intra-zone Routing:** Within each specific zone, the edge controller employs $Theta^*$ to identify the most efficient route between any two vehicles in the grid-based road network.
- **Inter-zone Routing:** For routing requests that span multiple zones, a global *SDN* controller uses a specialized version of $Theta^*$. It consults a centralized table containing routing information from all edge controllers, thereby extending route discovery capabilities beyond individual zones.

By leveraging $Theta^*$, *RADAR* achieves a high degree of routing efficiency and reliability, making it particularly well-suited for *SDN*-based *IoV* environments where rapid route discovery is essential.

3.2. RADAR methodology

In this subsection, we delve into the methodology that underpins *RADAR*, providing a detailed exploration of its algorithmic foundation, operational parameters, and data structures. Furthermore, we elaborate on the mechanics of *RADAR* in the contexts of both intra-zone and inter-zone route discovery. Leveraging *SDN* capabilities, *RADAR* efficiently discovers links/routes within individual zones using heuristic $Theta^*$, while also coordinating with global controllers for optimal route selection across multiple zones.

3.2.1. Grid representation of roads

To operationalize our algorithm, we represent the road network as a grid $G = (V, E, O)$ shown in Fig. 3, where:

- **Vertices** V: Comprising grid intersections at $V = \{v_{ij} \mid i = 1, 2, \ldots, m;\ j = 1, 2, \ldots, n\}$, it represents potential vehicular positions. The row and column indices range from 1 to m and 1 to n, respectively. Each vertex v_{ij} in the grid symbolizes a potential vehicular position within graph G.

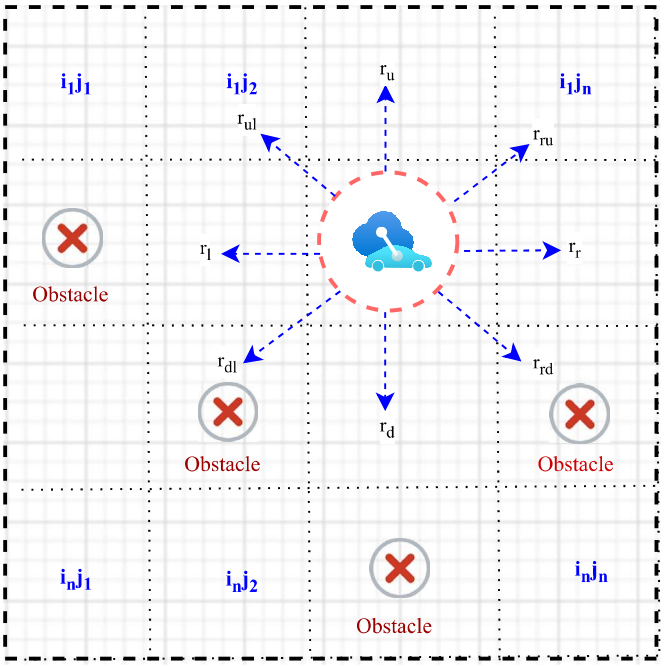


Fig. 3. Grid-based illustration of roads by *RADAR*.

- **Edges** E: Described by $E = \{(v_{ij}, v_{kl}) \mid |i-k| \leq 1, |j-l| \leq 1\}$ edges denote connections between adjacent vertices in specific grid cells. These connections can be oriented either horizontally, vertically, or diagonally.
- **Obstacles** O: Defined as $O = \{v_{ij} \mid M[i][j]\}$ obstacles are identified based on their presence in matrix M. Obstacles serve as barriers that could potentially affect vehicular communication reliability.
- **Link Reliability Function** f: We introduce a function $f : V \rightarrow LR$ that maps each vertex v to a set of link reliabilities LR. Formally, $f(v_{ij}) = \{r_u, r_d, r_l, r_r, r_{ul}, r_{dl}, r_{ur}, r_{dr}\}$ indicating the reliability of connections in 8 possible directions: up, down, left, right, and the 4 diagonals.

3.2.2. Matrix-based reliability mapping

We introduce a matrix MR, where each cell $MR[i][j]$ contains a set of link reliabilities LR for the corresponding vertex v_{ij}.

$$MR = \begin{pmatrix} LR_{11} & LR_{12} & LR_{1n} \\ LR_{21} & LR_{22} & LR_{2n} \\ LR_{m1} & LR_{m2} & LR_{mn} \end{pmatrix}$$

In this matrix, each LR_{ij} is a set $\{r_u, r_d, r_l, r_r, r_{ul} r_{dl}, r_{ur}, r_{dr}\}$, where each element represents the reliability of a link in a specific direction (up, down, left, right, up-left, down-left, up-right, down-right). These reliabilities are calculated based on Equation (5), which quantifies the robustness of a link. Matrix MR is derived from graph G and acts as a map, capturing the interconnectivity of vehicles in terms of link reliabilities. It is dynamically updated to reflect real-time changes, such as vehicle movements or the appearance and disappearance of obstacles. Within the *RADAR* framework, MR serves as the basis for all routing decisions. It is passed as an input to the $Theta^*$ algorithm to identify the most reliable paths. This approach facilitates rapid re-routing for link failures or other interruptions, effectively mapping the route from a specific source to a destination.

3.2.3. Intra-zone route discovery with RADAR

The $Theta^*$ algorithm is adopted in *RADAR* for rapid and reliable intra-zone routing via edge *SDN* controller. It aims to efficiently find a reliable path from a specific source to a destination in a given map MR, which represents the network topology. Algorithm 1, as given below, starts by initializing two sets: the open set, containing nodes to be evaluated, and the closed set, containing nodes that have already been evaluated. The cost function $g(u)$ for the source node is initialized to zero, and the source node is inserted into the open set along with its heuristic cost $h(u)$. The main loop of the algorithm continues as long as the open set is not empty. In each iteration, the node u with the highest reliability $g(u) + h(u)$ is removed from the open set. If this node is the destination, the algorithm terminates, returning the optimal route. Otherwise, u is added to the closed set, and its neighbors are considered for evaluation.

For each neighbor v of u, the algorithm checks whether v is in the closed set. If not, it proceeds to update v's costs and parent node based on whether a line-of-sight exists between v and the parent of u. The algorithm compares the new costs $g(v)$ against g_{old}. If $g(v)$ is less than g_{old}, indicating that a better path has been found, node v will be (re)inserted into the open set with its new costs. This ensures that the most reliable and efficient path is considered in subsequent iterations.

Algorithm 1 $Theta^*$ for rapid and reliable intra-zone routing in edge *SDN*.

Require: Map MR, Source vehicle sr, Destination vehicle ds
Ensure: Rapid and reliable rouing

```
Initialize open and closed sets: open := closed := ∅
g(sr) := 0
parent(sr) := sr
open.Insert(sr, g(sr) + h(sr))
while open ≠ ∅ do
    u := open.Pop()
    if u = ds then
        return route
    closed := closed ∪ {u}
    for each v in neighbors(u, MR) do
        if v ∉ closed then
            if v ∉ open then
                g(v) := ∞
                parent(v) := NULL
            g_old := g(v)
            if LOS(parent(u), v) then
                if g(parent(u)) + cost(parent(u), v) > g(v) then
                    parent(v) := parent(u)
                    g(v) := g(parent(u)) + cost(parent(u), v)
            else
                if g(u) + cost(u, v) > g(v) then
                    parent(v) := u
                    g(v) := g(u) + cost(u, v)
            if g(v) > g_old then
                if v ∈ open then
                    open.Remove(v)
                open.Insert(v, g(v) + h(v))
```

The worst-case time complexity for $Theta^*$ to discover the destination in a specific zone is $O(n \log n)$.

3.2.4. Inter-zone route discovery with RADAR (via global controller)

The algorithm deployed by the global controller uses a similar approach to $Theta^*$ but extends its functionality to work across multiple zones. The global controller maintains a comprehensive `globalRoutingTable` containing routing information from all edge controllers. Upon initialization, this table is populated to serve as a centralized repository for quick lookups.

As the algorithm continuously receives route discovery requests, each request is scrutinized to determine whether source s and destination d are within the same zone. If they are in the same zone, the task is delegated to the local Intra-Zone $Theta^*$ algorithm for route discovery, which has a time complexity of $O(n \log n)$. Conversely, if the source and destination are located in different zones, the global controller consults its `globalRoutingTable` to identify the edge controller responsible

Table 1
Simulation parameters and descriptions.

Parameter	Description
Simulation Tool	Matlab 2019b and *SUMO*
Network Topology	Grid-based road network
Grid Size	10×10
Road Length	2 km
Number of Lanes	2
RSUs Set	1 per street
SDN Edge Controller	2
RSU Coverage	2000 m
Traffic Signals	2
Vehicles' Communication Range	1000 m
Simulation Duration	500 s
Number of Vehicles	[10, 25, 50, 75, 100]
Vehicle Speed	Gaussian distribution
Routing Algorithms	*RADAR*, *Dijkstra*-based *SEARCH*, A^*-based
Scenario Types	2 (Sparse and Dense Obstacle)
Obstacle Density	Dense = 0.4; Sparse = 0.2

for the destination's zone. The route discovery request is then forwarded to this identified edge controller for localized and optimized route discovery.

This inter-zone forwarding ensures that each request, whether intra-zone or inter-zone, is processed with high efficiency, thereby maintaining the network's speed and reliability. The worst-case time complexity for this process, combining the search for an edge controller and the $Theta^*$ path-finding, is $O(n \log n)$.

Algorithm 2 $Theta^*$ Algorithm for Inter-Zone Route Discovery in Global Controller.

Input: globalRoutingTable, ZoneTable
Output: Most efficient route from source to destination
Initialize globalRoutingTable
Initialize requestQueue $= \emptyset$
while Controller is active **do**
  if Receive route request R with destination ds and source sr **then**
    requestQueue.Push(R)
  if requestQueue $\neq \emptyset$ **then**
    R = requestQueue.Pop()
    $sr = R$.source
    $ds = R$.destination
    if ZoneOf(ds, ZoneTable) = ZoneOf(sr, ZoneTable) **then**
      Perform Intra-Zone $Theta^*$ **Algorithm**
    else
      edgeController := Lookup(globalRoutingTable, ZoneOf(ds))
      Forward the request R **to edgeController**

4. Results and discussions

To evaluate the effectiveness of our *RADAR* framework, we conducted a series of experiments using MATLAB 2019b complemented by *SUMO* [32] as a mobility generator to produce random movements of vehicles. Initially, we established a grid network of roads and deployed various vehicles and obstacles at random locations. Next, we adopted a Gaussian distribution for the speeds of vehicles using *SUMO*, as mentioned earlier.

To assess the performance of our proposed *RADAR* framework, we compared it against relevant schemes, including *Dijkstra*-based *SEARCH* [16] and A^*-based route planning [33]. In this section, we first present the simulation model along with the various parameters considered for evaluation. Subsequently, we offer a detailed interpretation of the obtained results. The list of our parameters along with the description is shown in Table 1. We focused on the following three performance metrics:

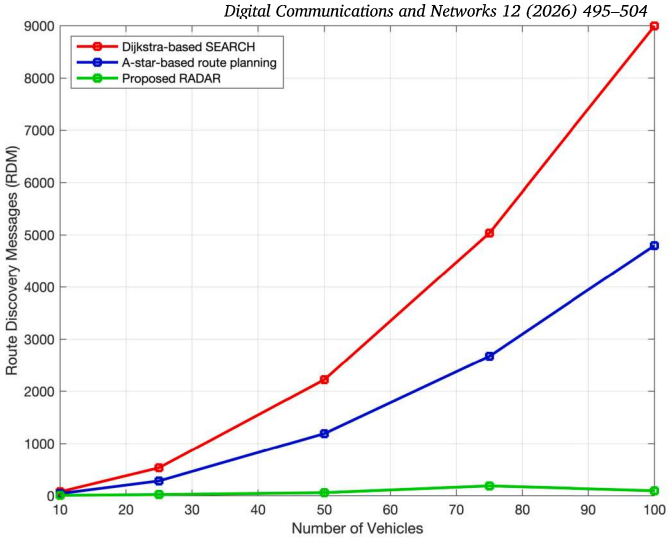


Fig. 4. Comparisons of Route Discovery Messages (*RDM*) across *Dijkstra*-based, A^*-based, and *RADAR* schemes in a dense obstacle environment. The *RADAR* scheme shows reduced *SDM*, highlighting its efficiency.

- **Route Discovery Messages (*RDM*):** This metric refers to the number of hello/beacon messages exchanged in the route discovery process from the source vehicle to the destination.
- **Average Path Length (*PL*):** This metric quantifies the average number of hops in the routes selected by all the compared schemes.
- **Route Discovery Time (*RDT*):** This measures the processing time required to find the optimal route from the source vehicle to the destination.

4.1. Performance of RADAR in dense obstacle environments

The obstacles pertain to the proportion of road segments that are non-traversable due to various blockages. The obstacle density of 0.4 represents a scenario with a higher concentration of such impediments, indicative of dense urban settings. In this subsection, we evaluate the performance of our proposed *RADAR* scheme in a dense obstacle environment, focusing on key performance metrics: *RDM*, *PL*, and *RDT*. Fig. 4 provides a comparative analysis of *RADAR* against A^*-based route planning [33] and *Dijkstra*-based *SEARCH* [16] for *RDMs*.

Our findings indicate that *RADAR* outperforms the other schemes, particularly in reducing the number of *RDMs*. This superior performance is attributed to *RADAR's* utilization of LOS for path planning, thereby minimizing the need for additional messages. Additionally, we observe that the number of vehicles in the network has a discernible impact on *RDMs* for all schemes. However, *RADAR* is significantly less affected, with the impact ranging from 18 to 97 fewer messages than *Dijkstra* and A^*-based approaches, as shown in Fig. 4.

Among the traditional algorithms, A^* exhibits better performance than *Dijkstra*, which can be ascribed to its heuristic approach. *Dijkstra*-based *SEARCH*, by design, expands a greater number of boundaries during the route discovery process [34], leading to higher *RDM* counts. In terms of *RDM*, *RADAR* shows a striking mean improvement of approximately 80% over *Dijkstra* and about 70% over A^*. This inherent characteristic of *Dijkstra's* algorithm makes it less efficient in dense obstacle environments when compared to both A^* and *RADAR*.

In addition to *RDMs*, we also evaluate the effectiveness of our proposed *RADAR* scheme in terms of *PL*. As depicted in Fig. 5, *RADAR* consistently identifies routes with fewer hops compared to A^* [33] and *Dijkstra*-based algorithms [16]. This efficiency can be attributed to *RADAR's* diagonal discovery approach and LOS check, which streamlines the route-finding process.

The density of vehicles in the network also plays a role in influencing *PL* across all schemes. As the number of vehicles increases, there is a cor-

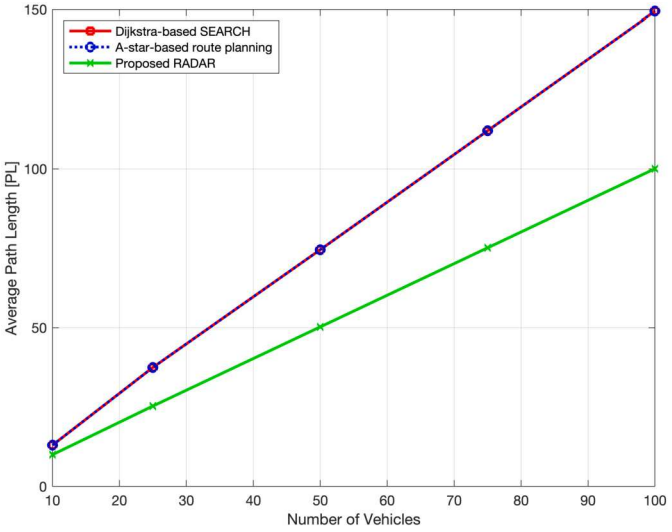


Fig. 5. Comparative analyses of average Path Length (*PL*) using *Dijkstra*-based, A^*-based, and *RADAR* schemes in a dense obstacle environment. *RADARv* demonstrates competitive path lengths, underscoring its efficacy. *Dijkstra*-based and A^*-based schemes are overlapped.

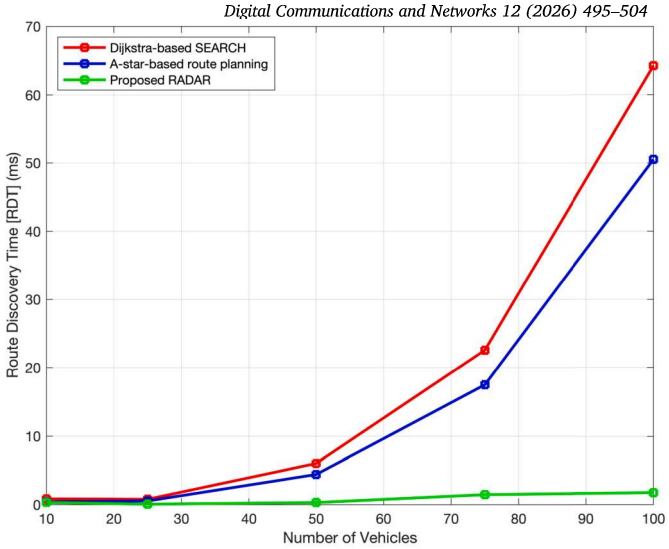


Fig. 6. Comparative evaluation of estimated Route Discovery Time (*RDT*) in milliseconds for *Dijkstra*-based, A^*-based, and *RADAR* schemes. The plot reveals *RADAR's* efficiency in terms of route discovery time.

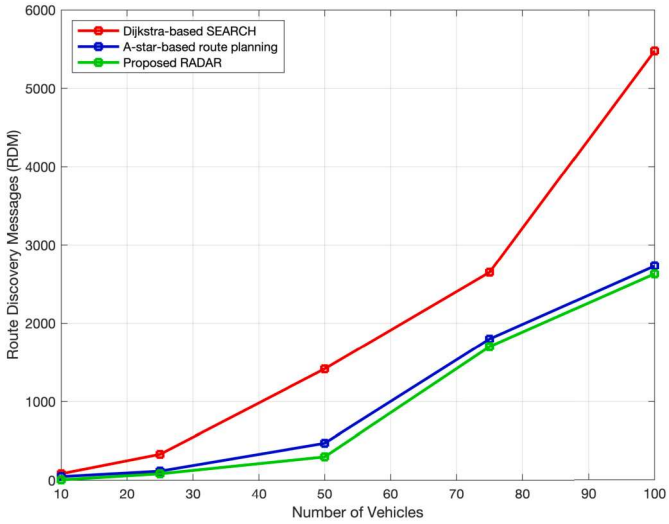


Fig. 7. Comparisons of Route Discovery Messages (*RDMs*) across *Dijkstra*-based, A^*-based, and *RADAR* schemes in a sparse obstacle environment. The *RADAR* scheme demonstrates a reduced number of *SDMs*, emphasizing its efficiency in such settings.

responding rise in *PL*; however, the effect is notably less pronounced in *RADAR* compared to A^* and *Dijkstra*-based methods. Intriguingly, both A^* and *Dijkstra* algorithms exhibit similar *PL* across different vehicle densities, confirming existing literature that these algorithms generally select equally optimal routes and albeit through different mechanisms [35]. A^*, thanks to its heuristic approach, manages to achieve this with fewer boundary expansions, making it more efficient than *Dijkstra* in this regard.

Our results underline *RADAR's* capacity for efficient route discovery and shorter path lengths, particularly in dense obstacle environments, as shown in Fig. 5. The *PL* is optimized considerably, with a mean improvement of 20% observed over both the *Dijkstra*-based *SEARCH* and A^* path planning schemes. This performance advantage validates the efficacy of *RADAR's* diagonal discovery approach and further sets it apart from traditional path planning algorithms like A^* and *Dijkstra*.

Finally, we focus on *RDT* in a dense obstacle environment, as illustrated in Fig. 6. Once again, *RADARv* exhibits exceptional performance, discovering optimal routes to destinations in just a few milliseconds. This rapid discovery can be largely attributed to *RADAR's* utilization of LOS techniques, which significantly reduce both *RDT* and *RDM* compared to other algorithms.

The number of vehicles in the network does have an impact on *RDT* across all considered schemes. However, even with increasing vehicle density, *RADAR* manages to maintain a discovery time of less than 5 milliseconds. This is notably superior to the times recorded for both *Dijkstra*-based and A^* path planning algorithms. As with the previous metrics, the A^* algorithm outperforms the *Dijkstra*-based approach. This advantage is consistent with our earlier discussions on *RDM* and *PL* and can be credited to A^*'s heuristic methods, which make it inherently more efficient.

In summary, our results strongly support the superiority of *RADAR* in terms of *RDT*, especially in dense obstacle settings, as shown in Fig. 6. In terms of *RDT*, *RADAR* continues to outperform, showing a mean improvement of around 70% and 60% over *Dijkstra* and A^*, respectively.

4.2. Performance of RADAR in sparse obstacle environments

Density 0.2 depicts a scenario with fewer obstacles, resembling sparser and less congested areas. In this subsection, we extend our performance evaluations to sparse obstacle environments, using the same metrics— *RDM*, *PL*, and *RDT*—, across *RADAR*, A^*-, based routing, and *Dijkstra-based SEARCH*.

In a sparse setting, Fig. 7 reveals a noticeable difference in the behavior of *RADAR* compared to its performance in dense obstacle environments. While *RADAR*'s *RDMs* are significantly reduced compared to the dense setting, its efficiency converges towards that of the A^* algorithm, albeit with a slight advantage. The impact of vehicle density on *RDMs* remains consistent with our earlier observations in the dense obstacle environment. Specifically, *RADAR* and A^* exhibit *RDMs* that vary between 20 and 2700, while the *Dijkstra* algorithm shows a much wider range, from 30 to 5500. Although *RADAR* maintains a slight edge over A^*, the performance gap has noticeably narrowed.

Turning to *PL*, as illustrated in Fig. 8, we observe that *PLs* vary from 18 to 125 for both A^* and *Dijkstra*-based schemes. *RADAR*, on the other hand, manages to maintain a slightly shorter path length, ranging from 15 to 120. While *RADAR* still demonstrates better performance in sparse environments, the degree of its advantage over traditional schemes is less pronounced compared to its performance in dense obstacle settings.

Finally, as depicted in Fig. 9, we analyze *RDT* in sparse obstacle conditions. Here again, *RDT* is considerably reduced across all schemes

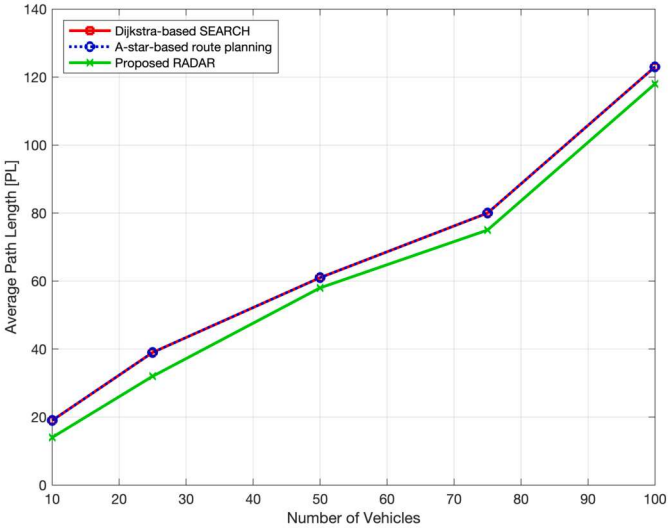


Fig. 8. Comparative analyses of Average Path Length (*PL*) using *Dijkstra*-based, A^*-based, and *RADAR* schemes in a sparse obstacle environment. *RADAR* demonstrates competitive path lengths, underscoring its efficacy. *Dijkstra*-based and A^*-based schemes are overlapped.

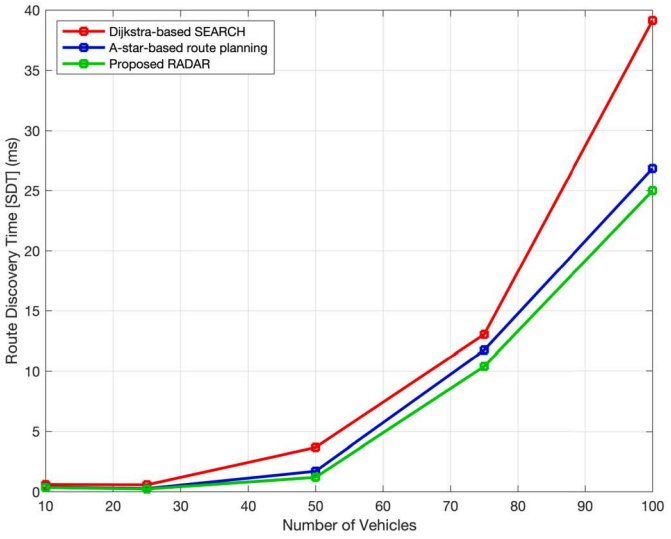


Fig. 9. Comparative evaluations of estimated Route Discovery Time (*RDT*) in milliseconds for *Dijkstra*-based, A^*-based, and *RADAR* schemes in a sparse obstacle environment. The plot reveals *RADAR*'s efficiency in terms of route discovery time.

compared to dense environments. *RADAR* demonstrates times ranging from 2 ms to 25 ms, A^* from 3 ms to 28 ms, and *Dijkstra* from 4 ms to 40 ms. Even though *RADAR* maintains its advantage, the margin of improvement is slimmer in sparse environments.

In summary, while *RADAR* retains its performance advantages in sparse obstacle environments, the extent of its superiority diminishes. Particularly in the metrics of *RDM* and *PL*, *RADAR's* edge narrows to be almost comparable with A^*. This suggests that while *RADAR* is highly effective in various settings, its performance gains are more dramatic in dense obstacle scenarios. The adaptability of *RADAR* to different obstacle densities is thus an area warranting further exploration.

4.3. Performance of RADAR in intra-zone and inter-zone environments

This subsection delves into the performance evaluation of our proposed *RADAR* algorithm, focusing on its operation in both inter-zone and intra-zone discovery scenarios. We apply the same performance

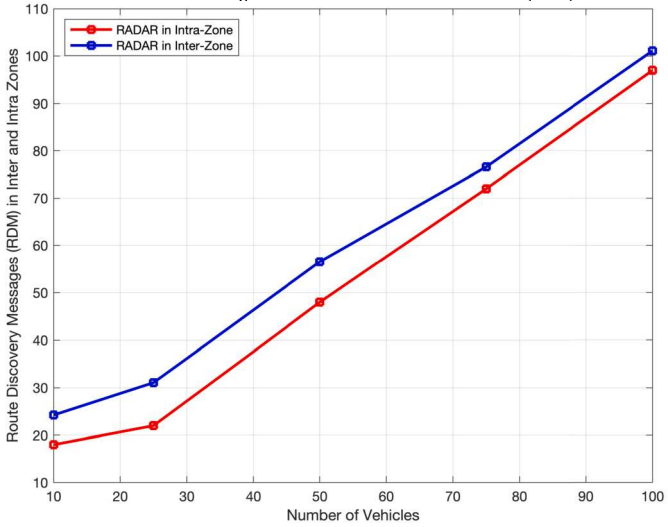


Fig. 10. Comparative analyses of Route Discovery Messages (*RDM*) of proposed *RADAR* under intra-zone and inter-zone settings.

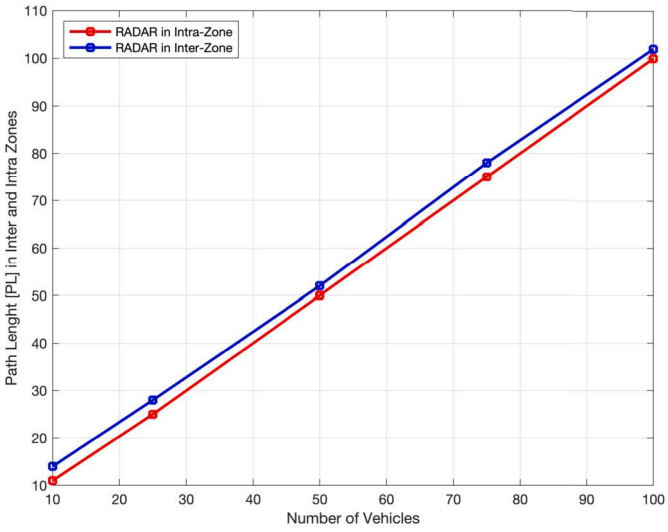


Fig. 11. Comparative analyses of estimated Path Length (*PL*) of proposed *RADAR* under intra-zone and inter-zone settings.

metrics used in previous sections—, namely, *RDM*, *PL*, and *RDT*—, and conduct these evaluations in a dense obstacle environment. The primary objective here is to examine how the edge and Global *SDN* controller influence the route discovery process in both zones.

Starting with *RDM*, Fig. 10 demonstrates that *RADAR* exhibits fewer *RDMs* in the intra-zone discovery process compared to its inter-zone counterpart. While the difference is not exceedingly large, it is still noteworthy. The marginally higher number of *RDMs* in the Inter-zone can be attributed to the necessity of accessing the Global controller, which subsequently consults its global routing table to identify the appropriate Edge *SDN* controller for the zone in question. In numerical terms, the *RDMs* for Intra-zone scenarios range from 20 to 95, while those for Inter-zone scenarios span from 24 to 100.

Next, we examine *PL* as depicted in Fig. 11. Here, we observe a subtle change between the *PL* values in inter-zone and intra-zone cases. The slight increase in *PL* for the inter-zone case is primarily due to the inclusion of an additional hop involving the global controller. Specifically, *PL* in intra-zone cases varies from 10 to 95; while in inter-zone cases, it ranges from 12 to 97.

Finally, as showcased in Fig. 12, we assess *RDT*. The variation between inter-zone and intra-zone scenarios is minimal but significant.

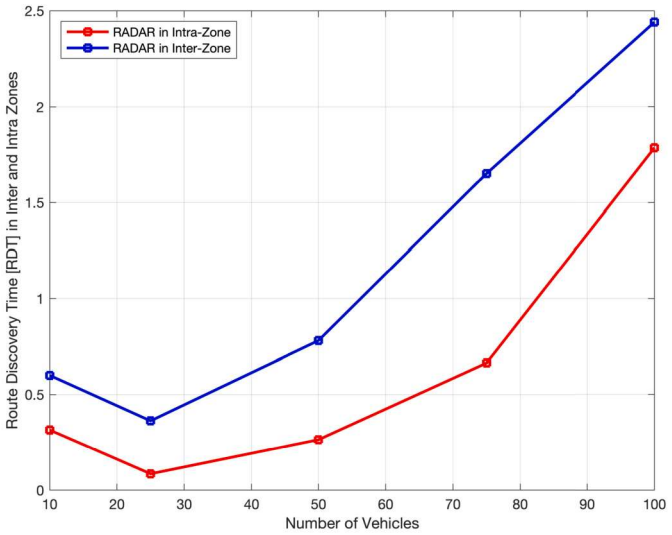


Fig. 12. Comparative analyses of Route Discovery Time (*RDT*) of proposed *RADAR* under intra-zone and inter-zone settings.

This minor uptick in *RDT* for the inter-zone case arises from the added latency involved in accessing the global controller and subsequently reaching another zone's edge *SDN* controller. In contrast, local discovery, which is restricted to the local edge *SDN* controller, proves to be less time-consuming. Quantitatively, the intra-zone discovery times fluctuate between 0.2 and 1.8 ms, while the inter-zone time ranges from 0.7 to 2.5 ms.

In summary, *RADAR's* local zonal (intra-zone) discovery consistently proves to be more efficient in terms of *RDM*, *PL*, and *RDT*, albeit with relatively minor margins. The additional latency and overheads in inter-zone discovery can be attributed to the involvement of the global controller, which validates *RADAR's* overall efficiency but also highlights areas for potential optimization, especially in multi-zone environments.

5. Conclusions

This paper introduces the *RADAR* algorithm, a robust and efficient scheme for rapid link and route discovery in *SDN*-based *IoV*. Utilizing a heuristic approach based on $Theta^*$, *RADAR* is integrated into *SDN* edge controllers to facilitate swift and efficient route discovery processes. Unlike conventional methods focusing solely on local discovery, *RADAR* broadens its reach by leveraging a global controller for expansive, inter-zone discovery while relying on edge *SDN* controllers for more localized and intra-zone operations. In terms of performance, *RADAR* stands out significantly across various metrics, i.e., *RDM*, *PL*, and *RDT*, when compared to state-of-the-art algorithms, such as Dijkstra-based search and A^* path planning. The empirical analysis reveals mean improvements of up to 80% in *RDM*, 20% in *PL*, and 70% in *RDT* over *Dijkstra*, as well as 70% in *RDM* and 60% in *RDT* over A^*. These findings collectively demonstrate that *RADAR* is an effective and promising solution for enhancing the reliability and efficiency of route discovery in complex *SDN*-based *IoV* environments. While the integration of *SDN* into the *IoV* framework introduces a degree of structural complexity, *RADAR*'s strategic application of *SDN* streamlines network management and significantly bolsters long-range communication capabilities, underscoring the trade-off as a justified and advantageous limitation of our approach. Our work opens up avenues for future research, including the potential integration of additional heuristics and the evaluation of *RADAR* under varying network conditions.

CRediT authorship contribution statement

Zahid Khan: Writing – original draft, Validation, Methodology, Investigation, Formal analysis, Conceptualization. **Nauman Khan:** Writing – review & editing, Visualization, Validation, Investigation, Formal analysis. **Anis Koubaa:** Writing – review & editing, Supervision. **Adel Ammar:** Writing – review & editing, Software. **Wadii Boulila:** Resources, Funding acquisition, Formal analysis.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

The work was supported by the Robotics and Internet-of-Things Lab of Prince Sultan University.

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